Title: Random Number Generation Using Mid-Square and Linear Congruential Methods with Testing by KS-Test

Objective

- ✓ To understand and implement random number generation using:
  - Mid-Square Method
  - Linear Congruential Method (LCG)
- ✓ To test the uniformity of the generated random numbers using the Kolmogorov–Smirnov (KS) Test.

Theory

2.1 Random Number Generation

Random numbers are numbers that occur in a sequence such that the values are uniformly distributed over a specified range and are independent of each other.

Pseudo-random numbers are generated using deterministic algorithms, which appear random but are completely reproducible given the same seed.

2.2 Mid-Square Method

- **Concept:**

  Start with a seed number, square it, extract the middle digits, and use them as the next seed.
- **Formula:**

$$X_{n+1} = \text{middle digits of } (X_n^2)$$

- **Normalized random number:**

$$R_n = \frac{X_{n+1}}{10^d}$$

  where *d* is the number of digits in the seed.
- **Example:**
  - Seed $X_0 = 5731$
  - $X_0^2 = 32844661$
  - Middle 4 digits = 8446 → $X_1 = 8446$
  - $R_1 = 8446/10000 = 0.8446$

2.3 Linear Congruential Generator (LCG)

- **Concept:**

  A linear recurrence relation generates a sequence of pseudo-random numbers.
- **Formula:**

$$X_{n+1} = (aX_n + c) \bmod m$$

$$R_n = \frac{X_{n+1}}{m}$$

- **Parameters:**
  - m: modulus
  - a: multiplier
  - c: increment
  - X_0: seed

2.4 Kolmogorov–Smirnov (KS) Test

Used to test whether the generated random numbers are **uniformly distributed** in [0,1).

- **Steps:**
  1. Sort the n generated random numbers $R_1, R_2, ..., R_n$ in ascending order.
  2. Compute for each i:

$$D^+ = \max_i \left(\frac{i}{n} - R_i \right)$$

$$D^- = \max_i \left(R_i - \frac{i-1}{n} \right)$$

  3. Calculate:

$$D = \max(D^+, D^-)$$

  4. Compare with the critical value:

$$D_\alpha = \frac{c(\alpha)}{\sqrt{n}}$$

    - For α = 0.05, $c(\alpha) = 1.36$
  - **If** $D < D_\alpha$: accept that numbers are uniform.
  - **Else:** reject uniformity.

Conclusion

• The experiment demonstrated two classic pseudo-random number generation techniques.
• The Mid-Square Method is simple but unstable for long sequences.
• The LCG Method provides better reproducibility and longer periods.
• The KS-Test confirmed the uniformity of the generated numbers.